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MPDMAC-SIC: Priority-based distributed low delay MAC with successive interference cancellation for multi-hop industrial wireless networks ☆

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ABSTRACT

Communications in industrial applications such as wireless factory automation demand different timing requirements. Providing timely medium access of the critical traffic and its prioritization over regular traffic at the same time is a significant challenge in industrial wireless networks. Successive Interference Cancellation (SIC) technique is an effective way to decrease access delay by allowing multiple packets reception concurrently. A series of novel Medium Access Control (MAC) protocols are proposed to differentiate access delay for various traffic types or exploit SIC for unique traffic type. Our protocol in this paper (MPDMAC-SIC) is the first priority based distributed MAC protocol that employs SIC to provide low access delay and accommodate different types of traffic for multi-hop industrial wireless networks. Two major contributions of our work are: first, we propose a low delay protocol for priority based multi-hop industrial wireless networks by enabling multiple packet reception on power domain with SIC. In MPDMAC-SIC, a power contention procedure other than traditional RTS/CTS contention in CSMA/CA DCF is introduced to allocate proper transmitting power for transmitters. Second, MPDMAC-SIC is modeled by discrete time Markov chain to minimize access delay by optimizing window size in power contention. According to the simulation results, MPDMAC-SIC performs better on access delay and packet arriving rate, compared to the existing good performing priority based CSMA/CA in both single-hop and multi-hop networks.

1. Introduction

Due to the lower cost and easier deployment, wireless networks for monitoring and control solutions have been applied widely in industry. In these industrial wireless networks, data traffic is divided into different categories: emergency traffic and regular traffic [1]. The category of traffic is assigned by industrial applications. Emergency traffic, e.g. control signal and accidental event signal, contains emergent information and thus needs to be transmitted faster than regular traffic. Therefore, priority system is essential to distinguish emergency traffic from regular traffic. High priority packets that contain emergency data demand low access delay and packet loss rate. The Medium Access Control (MAC) protocols that manage the access to the shared wireless channel are important to improve delay performance of emergency traffic. Because Carrier Sense Multiple Access/Collision Avoidance (CSMA/CA) provides low delay and is adapted for burst traffic in industrial wireless networks [2], MAC protocols based on CSMA/CA gain lots of attention.

Successive Interference Cancellation (SIC) is a promising technique to improve the performance of wireless networks by increasing spectrum efficiency [3]. The idea of SIC is to decode multiple signals sequentially by subtracting interference due to the decoded signals before decoding other signals. SIC relies on imbalance of the received powers from different signals transmitted by different transmitters.

In this paper, we propose a priority-based distributed low delay MAC scheme that adopts SIC (MPDMAC-SIC) for industrial wireless networks. SIC technique enables a node to receive signals from multiple transmitters simultaneously and thus improves the delay performance of network. In MPDMAC-SIC, high priority packets are allowed to transmit using either high power or low power, while low priority packets are only allowed to transmit using low power. Thus, high priority packets have more opportunities to access media than low priority packets. MPDMAC-SIC consists of two phases. The first phase is named RTS contention phase, which is a CSMA/CA like contention phase for transmitters to transmit Request To Send (RTS) packets. When an RTS packet reaches its destination, the second phase called power contention phase is triggered. Power contention phase gives transmitters that fail RTS contention an extra chance to transmit data. Winners of RTS contention phase and power contention phase are allowed to transmit data simultaneously. Because the transmitting power is regulated in the contention process, receiver can decode the mixed signals.

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MPDMAC-SIC is modeled by Markov chain and then the access delay is optimized in both single-hop and multi-hop networks. In performance evaluation, we show that MPDMAC-SIC performs better on access delay and packet arriving rate than the existing good performing priority assisted CSMA/CA.

The contributions of this paper are as follows:

(1) MPDMAC-SIC is the first priority based distributed MAC employing SIC, which improves access delay for industrial wireless networks.
(2) Performance of MPDMAC-SIC in both single-hop and multi-hop networks is modeled by Markov chain and then access delay is minimized.

The analytical model is verified by simulations and the results show that MPDMAC-SIC performs better than the existing priority assisted CSMA/CA.

The rest of this paper is organized as follows. Section 2 introduces the network model and MPDMAC-SIC. Performance of MPDMAC-SIC is analyzed in Section 3. Evaluation results are shown in Section 4, and conclusions are summarized in the last section.

2. Related works

MAC protocols, which determine the rules for nodes to access the wireless channel, greatly affect real-time performance of WSNs. Due to the difference on nodes of rule making, MAC protocols are divided into distributed protocols and centralized protocols. Distributed MAC protocols that are scalable and flexible are suitable for most WSNs. As a widely used distributed MAC protocol, CSMA/CA has been thoroughly studied. [4] modeled CSMA/CA to evaluate its performance, which considered many factors such as hidden terminal, interference and cell-overlapping. [5] proposed an algorithm to model CSMA/CA with Markov Chain and to derive the packet delays by calculating the probability distribution of stationary state of the Markov Chain. Above works lay a foundation for us to analyze the delay performance of distributed MAC protocols. The Enhanced Distributed Channel Access (EDCA) [6] was proposed to provide traffic differentiation for CSMA/CA by enabling high priority packets to access channel more frequently.

In recent years, many works that studied distributed MAC on traffic differentiation have been published. [7] combined reserving time slot and priority-based CSMA/CA to guarantee worst delay for emergent packets while improving average latency in the networks. [8] proposed a priority-based MAC protocol that adopted CSMA/CA for unmanned aerial vehicle WSN, which assigned smaller contention window size to transmission with higher priority. [9] proposed a hybrid node prioritization scheme based on CSMA/CA that reduced average channel access delay by contention window based prioritization. Based on the slotted CSMA/CA, [10] proposed an algorithm to reduce latency of real-time traffic, where nodes have different backoff range for packets with different priority. [11] assessed the performance of CSMA/CA that implemented the priority channel access mechanism. [12] improved QoS of EDCA by an adaptive contention window backoff mechanism. [13] proposed CSMA/Enhanced Collision Avoidance to provide traffic differentiation for wireless networks. Nevertheless, all these works allow at most one transmitter to access a given wireless channel at any instance, which limits the delay performance of the network.

SIC technique has been proposed more than thirty years, and it gains research interests again in recent years because of the popularity of NOMA (Non-orthogonal Multiple Access) technique in 5G [3, 14]. Due to its capability of multiple packets reception, SIC suggests a potential to improve channel utility and reduce delay. Many works have been carried out to study performance at wireless networks that employ SIC technique. [15–17] discuss how SIC affects CSMA/CA. [15] finds out that SIC technique is barely exploited by traditional CSMA/CA. [16] formulates the problem of determining the optimal probability distribution associating with power levels required by SIC. [17] presents a novel protocol based on multi-user information theory called CSMA k-SIC, which shows significant improvement compared to traditional CSMA/CA. [18] and [19] propose cross-layer optimization while employing SIC technique, involving physics, MAC and network layers. [18] jointly utilizes network coding and SIC to minimize the average delay of network. [19] develops a framework to study a network throughput maximization problem. [20] developed compact integer linear programming formulations to approach optimality of number of activating links that meet the SINR request of SIC. [21] proposed a mixed integer linear programming framework to maximize the rate that all links can simultaneously support. [22] presented a greedy method to manage the transmission concurrency based on SIC technique. [23] and [24] gave power optimized scheduling in guaranteed real-time networks. However, how to differentiate delay performance between regular traffic and emergency traffic remains an open problem while employing SIC in MAC protocols.

In [25], we proposed PDMAC-SIC to address the traffic differentiation with SIC technique. PDMAC-SIC employs SIC to provide low delay and accommodate different types of traffic for industrial wireless networks. An extra power contention procedure other than traditional RTS/CTS contention in CSMA/CA is introduced in PDMAC-SIC. This power contention procedure allows multiple transmitters to access the same channel simultaneously and thus access delay is decreased. PDMAC-SIC is modeled by Markov chain and then the access delay is minimized by optimizing the size of power contention window. Although PDMAC-SIC gave a solution to provide quite low delay for emergent traffics, it lacked analysis on multi-hop networks. To address this problem, we propose MPDMAC-SIC in this paper for adapting to applications in multi-hop networks.

3. Network model and algorithm description

We suppose that the network is constituted by a group of wireless nodes that share the same transmitting and receiving capabilities in this paper. Data traffic is divided into two categories: regular traffic and emergency traffic. Priority of data packets is assigned according to the traffic type, i.e. low priority packets for regular traffic and high priority packet for emergency traffic. Transmitters are able to switch their transmitting power between two levels. In this paper, transmitters are able to change their transmitting power in a given range. For the receiver, power difference between two signals that are received simultaneously need to satisfy equation (2). According to Eqs. (2) and (7), a transmitter can calculate its high transmitting power and low transmitting power when it is sending packets to the receiver. If this calculated low transmitting power is higher than the maximal transmitting power, this transmitter cannot reach the receiver in a single hop. Then this transmitter and the receiver are not neighbors. Otherwise, if the calculated low transmitting power is lower than the maximal transmitting power, the transmitter and the receiver are neighbors. Meanwhile, we assume that the transmitting rate is stable when the transmitting power is changed.

First, interference from non-neighbor nodes needs to be addressed. To simplify calculation, we use Signal to Interference plus Noise Ratio (SINR) model described in Eq. (1) to decide whether the signal can be decoded successfully.

$$SINR = \frac{P_r}{n_0 + P_I} \geq \gamma, \tag{1}$$

where γ is the threshold for correct reception of signals and the symbols of P_r, n_0 and P_I represent power of received signal, noise and interference, respectively.

In this network, SIC technique is implemented on receivers, so mixed signals can be decoded by the receivers if SINR of the signals meets Eq. (2).

$$\frac{P_r^H}{n_0 + P_I{}' + P_r^L} \geq \gamma, \qquad \frac{P_r^L}{n_0 + P'_I} \geq \gamma, \tag{2}$$

where P_r^H and P_r^L refer to power of received signal that are transmitted using high transmitting power and low transmitting power, respectively. The symbol $P_I{}'$ represents power of interference generated by other transmitting signals when SIC is employed in the network.

Only interference from non-neighbor nodes needs to consider since MAC algorithm addresses interference from neighbor nodes. Because all the nodes in this network transmit in the same channel, the power of interference signal is the summation of power from all non-neighbor nodes, as Eq. (3):

$$P_I = \sum_{i \in \bar{N}} P_r(i), \tag{3}$$

where $\bar{N}$ denotes the set of non-neighbor nodes. In MPDMAC-SIC process, only high priority packets have the opportunity to be transmitted using high power. Therefore, the proportion of high power signals in total signals is no more than the proportion of high priority packets in total packets. Let α denote the Proportion of High Priority Packets (PHPP). $P_I{}'$ is formulated as follows:

$$P_I{}' \le l \cdot \sum_{i \in \bar{N}} P_r(i) + \alpha \cdot k \cdot l \cdot \sum_{i \in \bar{N}} P_r(i), \tag{4}$$

where

$$k = \frac{P_r^H}{P_r^L} \tag{5}$$

denotes the lower bound of ratio of high received power to low received power, and

$$l = \frac{P_r^L}{P_r} \tag{6}$$

denotes the lower bound of ratio of low received power to received power without SIC.

Consequently, l is derived by P_r, n_0, γ according to Eqs. (1)–(6). For an instance, given $P_r = 2$, $n_0 = 1$, $\gamma = 1$, then $P_I = 1$, $l = 2/(2-\alpha)$ are derived. The result means that if PHPP $\alpha = 10\%$, the low received power when SIC is enabled is only 5% higher than the received power when SIC is disabled. With the propagation model of free-space:

$$P_r = d^{-\lambda} P_t \tag{7}$$

transmitting power is a linear function of receiving power, so the low transmitting power available for SIC is also only 5% more than transmitting power without SIC. When $P_r = 2$, $n_0 = 1$, $\gamma = 1$ are given, l can be obtained and then the lower bound of P_r^H can also be obtained according to Eqs. (2) and (6). With acquired P_r^L and P_r^H, interference from non-neighbor nodes is limited as the same as in the situation without SIC.

In this paper, MPDMAC-SIC is proposed to decrease access delay. All the control packets in MPDMAC-SIC are transmitted using low transmitting power. Process of MPDMAC-SIC consists with two phases. In the first phase, transmitters act as CSMA/CA: listening to the channel and contending channel by sending RTS. The first phase is named as "RTS contention phase". When the receiver receives an RTS, the second phase called "power contention phase" begins. An RTS for high priority packet triggers unrestricted power contention that allows an access from both high priority and low priority packets, while an RTS for low priority packet triggers restricted power contention that only allows an access from high priority packets. Restricted power contention and unrestricted power contention experience the same process except different contention window size and priority of packet allowed participating.

When employed in multi-hop networks, the protocol must face more complicated interference. For an instance, receiver in a single-hop network will only receive packets from transmitters, but in multi-hop networks it may also receive packets from other receivers. Therefore, two additional rules are added to MPDMAC-SIC to adapt multi-hop networks:

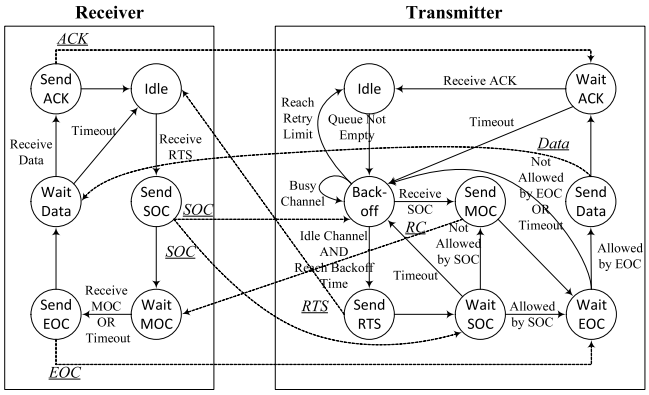


Fig. 1. Graph of state transition of MPDMAC-SIC.

(1) If a transmitter detects an EOC packet that is not from its target while waiting for SOC or EOC packet, it stops waiting and turns to the idle state.
(2) If a receiver detects an EOC packet while waiting for MOC packet, it stops waiting and turns to the idle state.

The process of MPDMAC is outlined as below. Fig. 1 gives a visual process of MPDMAC-SIC where solid arrows show state transitions and dash arrows show packet transmissions. Fig. 2 plots the process of power contention.

(1) When a transmitter has a data packet to send and detects an idle channel, it begins the RTS contention. The transmitter sends RTS packet after waiting random backoff time-slot. Different to CSMA/CA, RTS in MPDMAC-SIC contains the priority of data packet.
(2) When the receiver receives an RTS, it broadcasts a Start Of Contention (SOC) packet to its neighbors, which means that the power contention begins. If the receiver receives an RTS for high priority packet, an unrestricted power contention begins. Otherwise, a restricted power contention begins. Transmitters that have data packets with proper priority and have not sent their RTS are permitted to attend the power contention. Transmitters with high priority packets are allowed to attend both restricted power contention and unrestricted power contention, while transmitters with low priority packets are only allowed to attend unrestricted power contention.
(3) Transmitters that gain the permission from step 2 wait random time-slots selected in $[0, CW^{power}]$ and then send a Message Of Contention (MOC) packet to the receiver. CW^{power} is the general name for CW_{High}^{power} and CW_{Low}^{power} that are contention window sizes for restricted power contention and unrestricted power contention, respectively.
(4) The receiver broadcasts an End Of Contention (EOC) packet to its neighbors after receiving MOC. In the MOC, transmitting power is allocated as in Table 1.
(5) When the data is received successfully, the receiver sends an AC-Knowledgement (ACK) packet to the corresponding transmitters.

4. Analytical model of MPDMAC-SIC

In MPDMAC-SIC, power contention window size CW_{High}^{power} and CW_{Low}^{power} are key parameters that affect network performance. Because CW_{High}^{power} and CW_{Low}^{power} are calculated in the same way, CW^{power} is used as the general symbol for them to simplify description. In this paper, access delay represents the duration from the time that the data packet reaches the head of transmitting queue to the time that ACK of the data packet is received.

Table 1
Power Allocation Table.

Priority of packet wins RTS contention	Priority of packet wins power contention	Power for packet wins RTS contention	Power for packet wins power contention
High priority	None	Low power	None
High priority	High priority	Low power	High power
High priority	Low priority	High power	Low power
Low priority	None	Low power	None
Low priority	High priority	Low power	High power

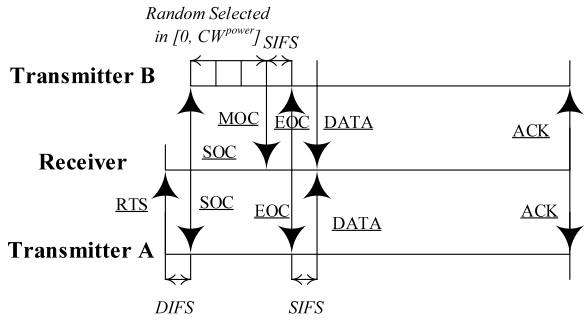


Fig. 2. Process of power contention of MPDMAC-SIC.

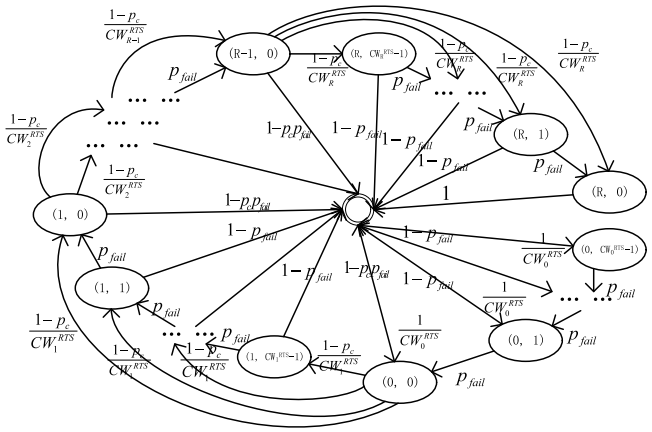


Fig. 3. Process of power contention of MPDMAC-SIC.

Let $r(t)$ and $b(t)$ be the stochastic processes that describe the number of retries of attempting to send RTS and the back-off timer for a given transmitter at slot time t, respectively. Markov chain that used to represent the process $\{r(t), b(t)\}$ is shown in Fig. 3. Let p_{fail}^{RTS} denote the probability that a transmitter sends an RTS and a collision occurs in a randomly given time slot during the RTS contention phase. Let p_{fail}^{power} denote the probability that a transmitter fails in the power contention. $p_{fail} = p_{fail}^{RTS} \cdot p_{fail}^{power}$ is the probability that a transmitter fails in both RTS contention phase and power contention phase.

Let $CW_r^{RTS} = 2^r \cdot CW_0^{RTS}$ denote size of the RTS contention window for a transmitter that has retried to send RTS for r times. The circle in the center of Fig. 3 represents the initial state of the transmitter, and state $s_{r,b}$ is the state that the transmitter has retried to send RTS for r times and its RTS contention back-off timer counts to b. To simplify the calculation, we only consider saturated traffic network in this paper. The stationary distribution of the Markov chain is:

$$p\left(s_{r,b}^*\right) = \lim_{t\to\infty} P\{r(t) = r, b(t) = b\}, \tag{8}$$

where $p\left(s_{r,b}^*\right)$ is derived from the matrix of state transition. Let p_{RTS} denote the probability that a transmitter sends a RTS packet in a time slot in the RTS contention phase. p_{RTS} is given by:

$$p_{RTS} = \sum_{r=0}^{R} p\left(s_{r,0}^*\right) \tag{9}$$

When all transmitters transmit RTS with the same probability, the probability of RTS collision is:

$$p_{fail}^{RTS} = 1 - \left(1 - p_{RTS}\right)^{n-1}, \tag{10}$$

where n denotes the number of transmitters in the contention. Let p_{tr}^{RTS} denote the probability that any RTS is transmitted in a random time slot, and p_{suc}^{RTS} denote the probability that an RTS is successfully transmitted in a random time slot and thus power contention is triggered. p_{tr}^{RTS} and p_{suc}^{RTS} are given by:

$$p_{tr}^{RTS} = 1 - \left(1 - p_{RTS}\right)^{n} \tag{11}$$

$$p_{suc}^{RTS} = \frac{n \cdot p_{RTS} \cdot \left(1 - p_{RTS}\right)^{n-1}}{1 - \left(1 - p_{RTS}\right)^{n}} \tag{12}$$

The average access delay is given by:

$$E[delay] = E\left[c_{slot}\right] \cdot E\left[d_{slot}\right], \tag{13}$$

where c_{slot} denotes the number of time slots in RTS contention phase required to transmit a packet successfully, and d_{slot} denotes the duration of one time slot in RTS contention phase. If no signal is listened by the transmitter, its back-off timer is not interrupted. Therefore, duration of an undisturbed time slot d_{slot}^u is just as the preset duration of time slot, i.e. $d_{slot}^u = T_{slot}$. Let d_{slot}^{con} denote duration of time slot when power contention is triggered, which is given by:

$$\begin{aligned} E\left[d_{slot}^{con}\right] &= T_{DIFS} + T_{RTS} + T_{SIFS} + T_{SOC} + E\left[d_{con}\right] \\ &+ T_{SIFS} + T_{EOC} + T_{SIFS} + T_{Data} + T_{SIFS} + T_{ACK} + T_{slot}, \end{aligned} \tag{14}$$

where $E\left[d_{con}\right]$ is the average duration of power contention phase. $T_{DIFS}, T_{SIFS}, T_{RTS}, T_{SOC}, T_{EOC}, T_{Data}$ and T_{ACK} are time needed for DIFS, SIFS, transmitting RTS, transmitting SOC, transmitting data and transmitting ACK, respectively. $E\left[d_{con}\right]$ is given by:

$$\begin{aligned} E\left[d_{con}\right] &= \sum_{i=1}^{R} p_{suc}^{power}\left(1 - p_{suc}^{power}\right)^{i-1} \\ &\cdot \left(\begin{array}{l} (i-1)\left(\begin{array}{l} CW^{power} + T_{DIFS} \\ +T_{RTS} + T_{SIFS} + T_{SOC} \end{array} \right) \\ +\frac{CW^{power}+1}{2}, \end{array} \right) \end{aligned} \tag{15}$$

where the probability that any transmitter wins in the power contention is:

$$\begin{aligned} p_{suc}^{power} &= \\ &\sum_{i=1}^{n-1} \frac{(-1)^{i-1} i! \left(\begin{array}{c} n \\ i \end{array} \right) \left(\begin{array}{c} CW^{power} \\ i \end{array} \right) (CW^{power} - i)^{n-i}}{(CW^{power})^n} \\ &+ \frac{(-1)^{n-1} n! \left(\begin{array}{c} CW^{power} \\ n \end{array} \right)}{(CW^{power})^n} \end{aligned} \tag{16}$$

Let d_{slot}^{fail} denote duration of time slot when no RTS is transmitted successfully and power contention is not triggered, which is given by:

$$E\left[d_{slot}^{fail}\right] = T_{DIFS} + T_{RTS} + T_{SIFS} + T_{slot} \tag{17}$$

The average duration of a time slot is:

$$E\left[d_{slot}\right] = \left(1 - p_{tr}^{RTS}\right) \cdot d_{slot}^{u} + p_{tr}^{RTS} \cdot p_{suc}^{RTS} \cdot E\left[d_{slot}^{con}\right] + p_{tr}^{RTS} \cdot \left(1 - p_{suc}^{RTS}\right) \cdot E\left[d_{slot}^{fail}\right] \tag{18}$$

The average number of time slot experienced for a successfully transmitted packet is:

$$E\left[c_{slot}\right] = \sum_{r=0}^{R} E\left[c_{slot}|r\right], \tag{19}$$

where $E\left[c_{slot}|r\right]$ denotes the average number of time slots experienced if the packet is successfully transmitted just after retry r times. $E\left[c_{slot}|r\right]$ is given as:

$$E\left[c_{slot}|0\right] = \frac{\sum_{b=0}^{CW_0^{RTS}} b \cdot p\left(s_{0,b}^{*}\right)}{\sum_{b=0}^{CW_0^{RTS}} p\left(s_{0,b}^{*}\right)} \tag{20}$$

$$E\left[c_{slot}|r\right] = \sum_{i=0}^{r-1} E\left[c_{slot}|i\right] + \frac{\sum_{b=0}^{CW_r^{RTS}} b \cdot p\left(s_{r,b}^{*}\right)}{\sum_{b=0}^{CW_r^{RTS}} p\left(s_{r,b}^{*}\right)}, r > 0 \tag{21}$$

The relation between $E[delay]$ and $p\left(s_{r,b}^{*}\right)$ is derived from Eqs. (13)–(21). Then, from Eqs. (8), the optimized CW^{power} to get the minimum access delay is calculated.

5. Extensive analysis in multi-hop networks

In the previous section, an analytical model for MPDMAC-SIC in single-hop networks is proposed to calculate access delay and get the optimized contention window size. In this section, we will calculate the access delay in multi-hop networks. The difference of access delay between single-hop and multi-hop networks comes from the interference from receivers other than the target of a transmitter. In order to analyze the effect generated by such interference, probability distribution of stationary state of the receivers is modeled.

To simplify the calculation, we assume that each receiver has the same number of neighbored transmitters. According to Fig. 1, states of a receiver are divided into three categories:

(1) Idle state. This state contains "idle" state in Fig. 1. In the idle state, the receiver waits for RTS packets from transmitters and does not transmit any control packets.
(2) Contention state. This state contains three states of "send SOC", "wait MOC" and "send EOC" in Fig. 1. In the contention state, the receiver processes RTS packets from transmitters determines the result of contention. The EOC packet is transmitted in this state.
(3) Receiving state. This state contains "wait Data" and "send ACK" state in Fig. 1. In the receiving state, the receiver receives data and transmits acknowledge packet that does not affect other transmitters.

Let p_{rij} denote the probability of state transiting probability of a receiver. For example, p_{r12} is the probability that a receiver transits its state from the idle state to the contention state. As the result, state transitions of receivers in MPDMAC-SIC are illustrated as Fig. 4.

Meaning of symbols in Fig. 4 is described below. p_{RTS} is as mentioned in the previous section, which is the probability that a transmitter sends an RTS packet in a time slot. p_{EOC} is the probability that a receiver sends an EOC packet in a time slot. n and n' are number of neighbored transmitters and receivers, respectively.

Let $\{\pi_{r1}, \pi_{r2}, \pi_{r3}\}$ denote the probability of stationary state of the receivers. From Fig. 4, $\pi_{r1}, \pi_{r2}, \pi_{r3}$ can be derived. Because $p_{r12} = p_{RTS}\left(1 - p_{RTS}\right)^{n}$ and $p_{r23} = p_{EOC}\left(1 - p_{EOC}\right)^{n}$, we have:

$$\pi_{r1} = \frac{1}{2 + p_{12} + p_{12}p_{23}} \tag{22}$$

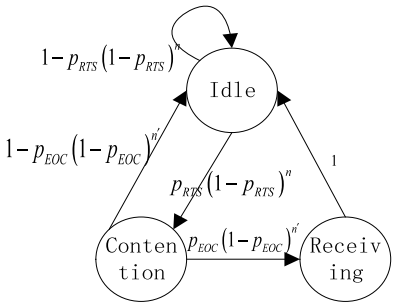


Fig. 4. Graph of state transition of receivers.

$$\pi_{r2} = \frac{p_{12}}{2 + p_{12} + p_{12}p_{23}} \tag{23}$$

$$\pi_{r3} = \frac{1 + p_{12}p_{23}}{2 + p_{12} + p_{12}p_{23}} \tag{24}$$

Then p_{EOC} can be derived:

$$p_{EOC} = \frac{T_{r3} \cdot \pi_{r3}}{T_{r1} \cdot \pi_{r1} + T_{r2} \cdot \pi_{r2} + T_{r3} \cdot \pi_{r3}} \cdot \frac{T_{EOC}}{T_{r3}}, \tag{25}$$

where T_{r1}, T_{r2}, T_{r3} represent average duration when a receiver stays in idle state, contention state and receiving state respectively. T_{EOC} is the time needed to send an EOC packet.

Let p_t^{EOC} denote the probability that a transmitter receives an EOC packet not from its target when the transmitter is waiting SOC or EOC packets. From above equations, p_t^{EOC} is derived:

$$p_t^{EOC} = \frac{\left(T'_{SOC} + T'_{EOC} + T_{EOC}\right) \cdot p_3}{T_{r1} \cdot \pi_{r1} + T_{r2} \cdot \pi_{r2} + T_{r3} \cdot \pi_{r3}}, \tag{26}$$

where T'_{SOC} and T'_{EOC} represents the average duration when a transmitter waits for an SOC packet and an EOC packet respectively.

Let $E\left[delay_{suc}|r\right]$ denote the average duration of a retry when a transmitter retries r times and succeeds. Let $E\left[delay_{fail}|r\right]$ denote the average duration of a retry when a transmitter retries times and failures. $E\left[delay_{suc}|r\right]$ and $E\left[delay_{fail}|r\right]$ are given by:

$$E\left[delay_{suc}|r\right] = E\left[c_{slot}|r\right] \cdot E\left[d_{slot}\right] \tag{27}$$

$$E\left[delay_{fail}|r\right] = E\left[delay_{suc}|r\right] - T_{SIFS} - T_{EOC} - T_{SIFS} - T_{Data} - T_{SIFS} - T_{ACK} \tag{28}$$

In conclusion, the average access delay in multi-hop networks is:

$$E\left[delay_{mul}\right] = \left(1 - p_t^{EOC}\right) \sum_{i=1}^{r} \left(p_t^{EOC}\right)^{i-1} E\left[delay_{suc}|r\right] + \left(1 - p_t^{EOC}\right) \sum_{i=1}^{r-1} E\left[delay_{fail}|r\right] \sum_{j=1}^{r-i} \left(p_t^{EOC}\right)^{j} \tag{29}$$

6. Performance evaluation

The industrial wireless network is simulated using the discrete event-driven network simulator WSNet [26]. The network is composed of one receiver and several transmitters random distributed around the receiver as Fig. 5. This star topology is typical in single-hop industrial wireless networks [27]. Traffic of the network is saturated. Category of traffic is assigned randomly with certain probability. Therefore the proportion of emergency traffic is fixed in each simulation. Other system parameters in the simulation are shown in Table 2, where the max transmitting power and receiving power are from the mica2 model [28].

Table 2
Power Allocation Table.

Parameter	Value	Parameter	Value
Area size	200 m × 200 m	Maximal transmitting power	21.5 mW
Number of transmitters	10 to 20	Transmitting power tuning step	1 mW
Radio propagation model	Free space	Reception power	7 mW
Battery model	Linear	CW_0^{RTS}	8
Number of data packets	10000	R	7
Size of data packet	10000 bit	T_{SIFS}	10 μs
Size of control packet	168 bit	T_{DIFS}	60 μs
Bit rate	1.38 Mb/s	T_{slot}	20 μs

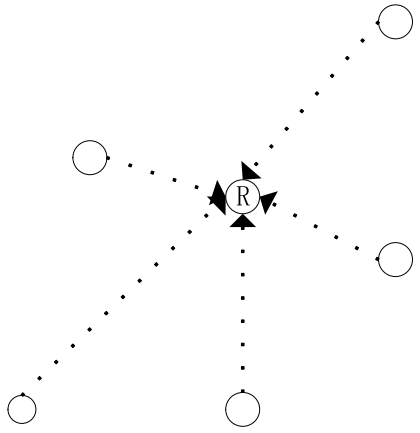


Fig. 5. Topology of single-hop network in the simulation.

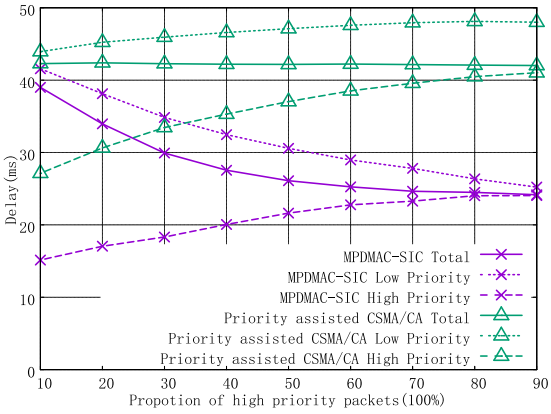
The results, i.e. access delay, packet arriving rate and energy consumption, are compared to Priority assisted CSMA/CA [9] in the simulation. The low priority traffic is "Best effort" type and the high priority traffic is "Voice" type. Contention window size for different traffic types in the priority assisted CSMA/CA is listed in [9]. Each simulation runs for 100 times.

- Access Delay

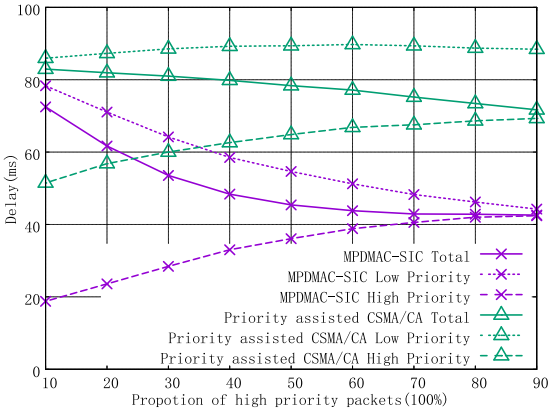
  Fig. 6 indicates that MPDMAC-SIC reduces access delay of high priority packets effectively. On average, access delay of high priority packets in MPDMAC-SIC is about 30% shorter than in the compared protocol. In both MPDMAC-SIC and the compared protocol, delay of high priority packets is shorter than delay of low priority packets. However, the mechanics to provide shorter delay for high priority packets are different.

  In the compared protocol, smaller contention windows are assigned to high priority packets. Thus a transmitter with high priority packet is more likely to select a smaller time slot during contention and more likely to win a contention. When proportion of high priority packet (PHPP) is low, both high priority and low priority packets achieve decent access delay in the compared protocol. However, with the increasing of PHPP, access delay of all packets increases rapidly. For an instance, when PHPP increases from 10% to 50% in the network with 10 transmitters, access delay of high priority packets increases about 10 ms and access delay of low priority packets increases about 4 ms. The reason is PHPP is high, more transmitters contend with smaller contention window, which leads to more collisions. As the result, both high priority packets and low priority packets suffer longer access delay when PHPP is high. The smaller contention window also lengthens access delay when number of transmitters increases. According to Fig. 6(c), when number of transmitters increase from 10 to 20, average access delay and access delay of high priority packets increases by about 30 ms.

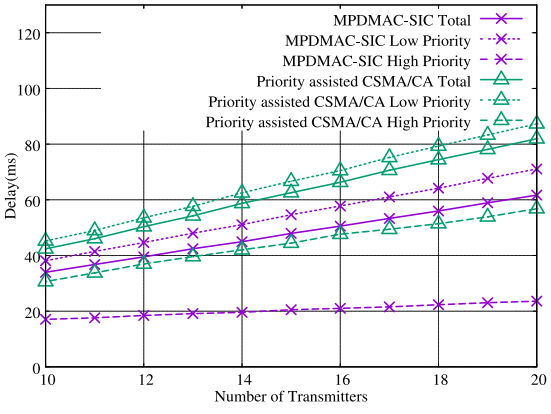
  In MPDMAC-SIC, access delay of high priority packets is about 50% shorter than in the compared protocol. Winners of RTS



(a) 10 transmitters



(b) 20 transmitters



(c) Network size increases (PHPP = 20%)

Fig. 6. Comparison of access delay.

contention and power contention are allowed to transmit concurrently in MPDMAC-SIC. Thus channel utility is improved and access delay is shortened. In the RTS contention, both high priority packets and low priority packets are treated equally. But in the power contention, high priority packets are allowed to participate in both restricted power contention and unrestricted power contention, while low priority packets are only allowed to participate in unrestricted power contention. In this manner, high priority packets have more chance win a contention. In MPDMAC-SIC, access delay is relative stable when PHPP changes. When PHPP increases from 10% to 50% in the network with 10 transmitters, access delay of high priority packets increases only 6 ms while access delay of low priority packets also decreases 11 ms. When the most of packets are low priority packets, most RTS contentions are won by low priority packets. RTS for low priority packet triggers restricted power contention that only allows high priority packets to contend. Since high priority packets are relative few, a high priority packet probably wins the first accessed restricted power contention and thus achieves quite low access delay. Increase of PHPP causes more unrestricted power contentions triggered, which gives an extra chance to both low priority and high priority packets for parallel transmission. Therefore the channel utility is improved and the access delay remains stable when PHPP increases. For the same reason, access delay changes slightly when number of transmitter grows. When number of transmitters increases from 10 to 20, access delay of high priority packets increases only 6 ms, which is 80% less than in the compared protocol.

- Packet Arriving Rate

  Fig. 7 plots different Packet Arriving Rate (PAR) in MPDMAC-SIC and in the compared protocol.

  In the compared protocol, smaller contention window for high priority packets causes more intense contention, which makes more packets be disposed after reaching retry limit of transmission. So PAR of high priority packets is rather low. When the size of network grows, PAR of high priority packets decreases rapidly. For example, when there are 20 transmitters in the network, PAR of high priority packets is only about 80%, while PAR of low priority packet is about 99%.

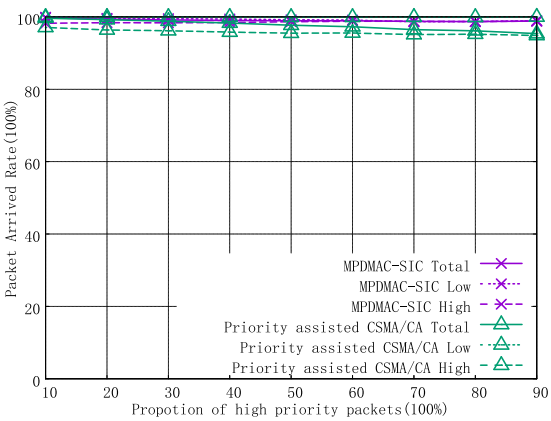
  In MPDMAC-SIC PAR almost remains the same when PHPP and scale of network changes. Different from the compared protocol, low priority and high priority packets are treated equally in RTS contention phase. In addition, high priority packets that are able to participate in restricted power contentions have better chance to be transmitted before reaching the limit of retry. Because of higher channel utility, MPDMAC-SIC improves access delay without deteriorating PAR for high priority packets.

- Energy Consumption

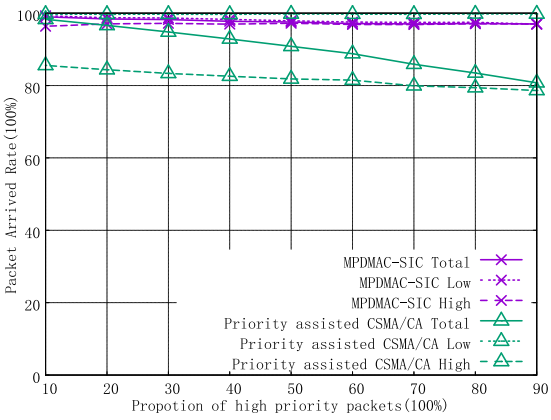
  Fig. 8 shows that power consumption in MPDMAC-SIC is higher than in the compared protocol. The reason is that SIC technique requests more power for transmitting signals. When 90% of the packets are high priority packets, MPDMAC-SIC consumes about 20% more energy than the compared protocol in the network with 10 transmitters. Meanwhile, when 10% of the packets are high priority packets, MPDMAC-SIC consumes only 8% more energy.

The results indicate that MPDMAC-SIC exploits SIC technique effectively. Compared to conventional priority assisted CSMA/CA, MPDMAC-SIC performs much better on both access delay and PAR at the expense of a little more energy consumed.

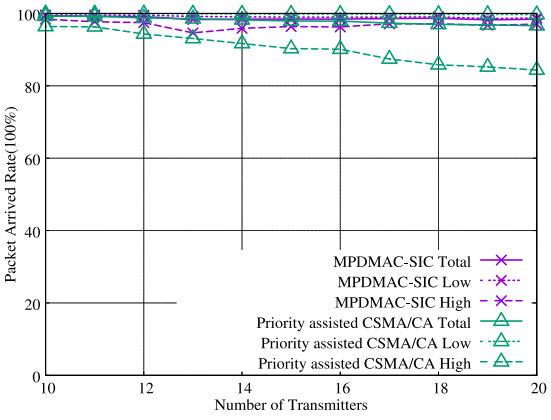
The simulation of MPDMAC-SIC in multi-hop networks is demonstrated below. Network as Fig. 9, which is typical in industry wireless networks [29], is used in this simulation. In this simulation, node 0 is the sink node, which only receives data from other nodes. Nodes 1, 2, 3, 6, 9, 10 are source nodes, which only transmit data to other nodes. Nodes 4, 5, 7, 8 are relay nodes, which receive data from upstream



(a) 10 transmitters



(b) 20 transmitters



(c) Network size increases (PHPP = 20%)

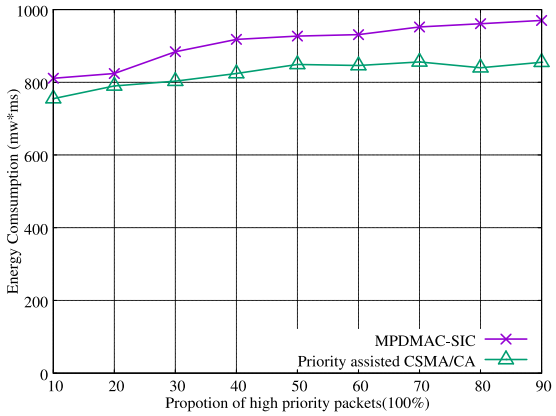
Fig. 7. Comparison of packet arriving rate.

nodes and then transmit it to downstream nodes. Data packets are generated in the source nodes and sent to the sink node. The other parameters are as the same as listed in the previous section.

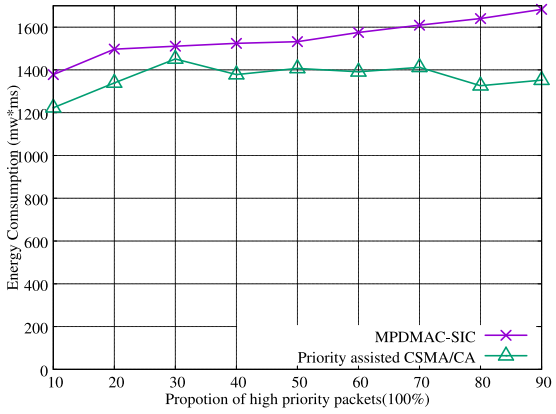
The results of the simulations in multi-hop networks are illustrated in Figs. 10–12.

- Access Delay

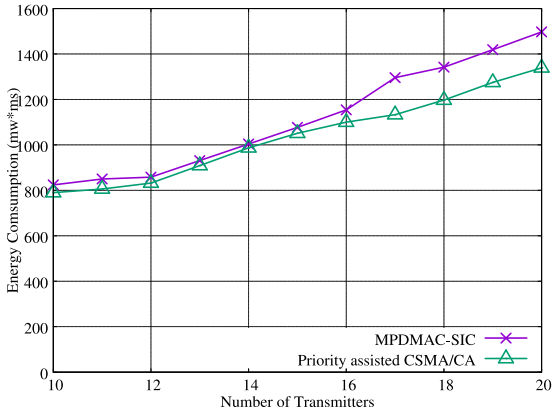
  Fig. 10 indicates that MPDMAC-SIC performs well to reduce access delay in multi-hop networks. Average access delay in MPDMAC-SIC is about 20% shorter than in the compared protocol. However, different from single-hop network, when PHPP changes, average access delay in both MPDMAC-SIC and the compared protocol merely change. The reason comes from the



(a) 10 transmitters



(b) 20 transmitters



(c) Network size increases (PHPP = 20%)

Fig. 8. Comparison of energy consumption.

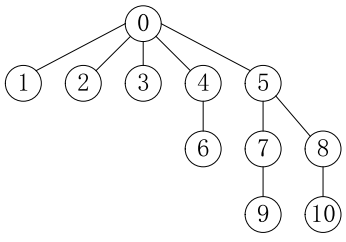


Fig. 9. Topology of multi-hop network in the simulation.

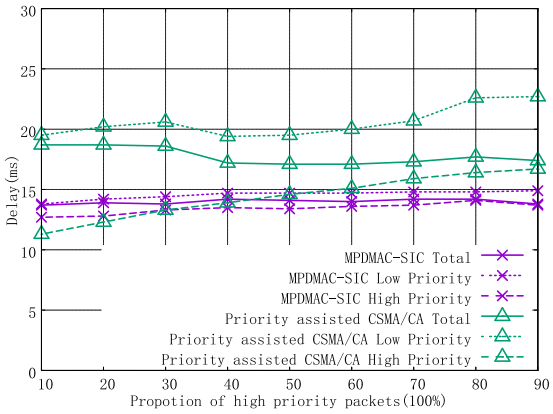


Fig. 10. Comparison of access delay.

topology of this multi-hop network. Relay nodes 4, 7 and 8 receive data from single transmitter. Only sink node 0 and relay node 5 receive signals from multiple transmitters.

This type of topology reduces contentions compared to the star type single-hop network. This alleviates collisions caused by short window size for high priority packets in the compared protocol. As the result, when PHPP is low, high priority packets in both MPDMAC-SIC and the compared protocol are able to access to the channel within a short time. Nevertheless, contentions in node 0 and 5 are inevitable. When PHPP increases from 10% to 90%, access delay of high priority packets increases from 12 ms to 17 ms, and access delay of low priority packets increases from 20 ms to 23 ms.

Since MPDMAC-SIC cannot make multiple signals be transmitted concurrently to relay nodes 4, 7, 8, advantage of SIC is weakened. Access delay changes only 2 ms when PHPP increases from 10% to 90%. This results show that MPDMAC-SIC outperform the priority-assisted CSMA/CA on the aspect of access delay in a multi-hop network. Together with the results from the single-hop network, the conclusion is that MPDMAC-SIC reduces access delay more effectively in the network with multiple transmitters sending packets to one receiver.

- Packet Arriving Rate

  Fig. 11 compares PAR between the two protocols. MPDMAC-SIC achieves better PAR of high priority packets regardless the PHPP. Furthermore, PAR in MPDMAC-SIC keeps high when PHPP changes, while PAR in the compared protocol decrease rapidly when PHPP increases. The result indicates that the compared protocol acquires low access delay with the cost of disposing packets frequently, and this disadvantage still exists in multi-hop networks.

- Energy Consumption

  Fig. 12 plots the energy consumption against PHPP in the multi-hop network. Similar as in the single-hop network, energy consumption in MPDMAC-SIC increases as PHPP increases. When PHPP is 10%, energy consumptions in MPDMAC-SIC and in the compared protocol are almost the same. However, when PHPP is 90%, MPDMAC-SIC consumes 18% more energy than the compared protocol.

In conclusion, MPDMAC-SIC achieves lower access delay and higher PAR with the cost of energy consumption. For an instance, when PHPP is 50%, access delay and PAR in MPDMAC-SIC are respectively 30% shorter and 12% higher than in the compared protocol. But energy consumption in MPDMAC-SIC is 10% more than in the compared protocol.

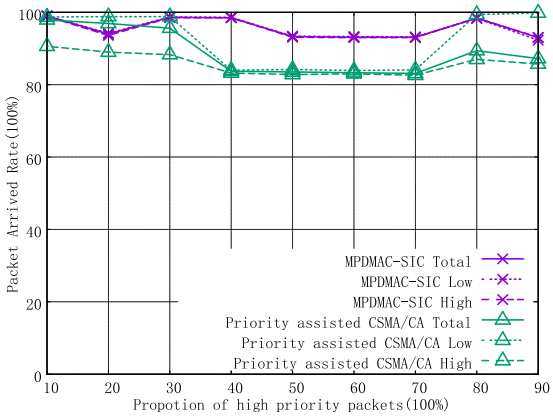


Fig. 11. Comparison of packet arriving rate.

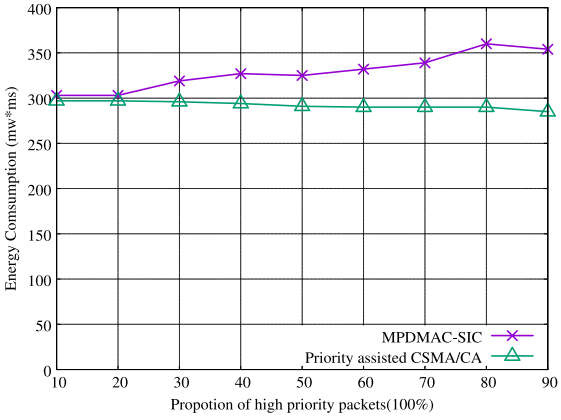


Fig. 12. Comparison of energy consumption.

7. Conclusion

In this paper, we propose a priority-based distributed low delay MAC scheme that employs SIC (MPDMAC-SIC) for industrial wireless networks. MPDMAC-SIC is the first MAC protocol that provides better service for high priority packets while enables SIC to reduce access delay in multi-hop network. MPDMAC-SIC provides opportunities for parallel transmissions by introducing additional contention phase named "power contention phase". Because of higher channel utility, MPDMAC-SIC improves access delay without deteriorating packet arriving rate for high priority packets. Compared to the existing priority assisted good performing CSMA/CA, MPDMAC-SIC performs better in access delay and packet loss rate with the trade-off of limited energy consumption. This work gives a method to integrate priority differentiation and power-domain NOMA, which can be further improved to adapt to more complex environment.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Xu Yida: Conceptualization, Methodology, Investigation, Visualization, Writing - original draft. **Wang Qi:** Supervision, Writing - review & editing. **Xu Yongjun:** Project administration. **Liu Jianmin:** Writing - review & editing. **He Chentao:** Writing - review & editing.

References

[1] I. Standard, Wireless systems for industrial automation: process control and related applications, ISA-100.11 a-2009, 2009.
[2] V. Bharghavan, A. Demers, et al., MACAW: a media access protocol for wireless LAN's, ACM SIGCOMM Comput. Commun. Rev. 24 (4) (1994) 212–225.
[3] S. Islam, N. Avazov, et al., Power-domain non-orthogonal multiple access (NOMA) in 5G systems: Potentials and challenges, IEEE Commun. Surv. Tutor. 19 (2) (2016) 721–742.
[4] K.C. Huang, K.C. Chen, Interference analysis of nonpersistent CSMA with hidden terminals in multicell wireless data networks, in: Proceedings of 6th International Symposium on Personal, Indoor and Mobile Radio Communications, 2002.
[5] P. Chatzimisios, IEEE 802.11 packet delay-a finite retry limit analysis, 2009.
[6] IEEE 802.11 Working Group, et al., IEEE Standard for Information Technology—Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications Amendment 8: Medium access control (MAC) quality of service enhancements, IEEE Standards Association, Piscataway, NJ, USA, 2005.
[7] M. Collotta, L. Gentile, G. Pau, G. Scata, Flexible IEEE 802.15.4 deadline-aware scheduling for DPCSs using priority-based CSMA-CA, Comput. Ind. 65 (8) 1181–1192.
[8] S. Sotheara, N. Aomi, T. Ando, L. Jiang, N. Shiratori, S. Shimamoto, Effective data gathering protocol in WSN-UAV employing priority-based contention window adjustment scheme, in: 2014 IEEE Globecom Workshops, GC Wkshps, IEEE, 2014, pp. 1475–1480.
[9] M. Shakir, O. Rehman, et al., Performance optimization of priority assisted CSMA/CA mechanism of 802.15. 6 under saturation regime, Sensors 16 (9) (2016) 1421.
[10] I. Dbibih, I. Iala, O. Zytoune, D. Aboutajdine, Message priority CSMA/CA algorithm for critical-time wireless sensor networks, in: Proceedings of the Mediterranean Conference on Information & Communication Technologies 2015, Springer, 2016, pp. 265–274.
[11] L. Leonardi, G. Patti, F. Battaglia, L.L. Bello, Simulative assessments of the IEEE 802.15. 4 CSMA/CA with priority channel access in structural health monitoring scenarios, in: 2017 IEEE 15th International Conference on Industrial Informatics, INDIN, IEEE, 2017, pp. 375–380.
[12] I. Syed, S.-h. Shin, B.-h. Roh, M. Adnan, Performance improvement of QoS-enabled WLANs using adaptive contention window backoff algorithm, IEEE Syst. J. 12 (4) (2017) 3260–3270.
[13] L. Sanabria-Russo, B. Bellalta, Traffic differentiation in dense collision-free WLANs using CSMA/ECA, Ad Hoc Netw. 75 (2018) 33–51.
[14] L. Lei, D. Yuan, C.K. Ho, S. Sun, Power and channel allocation for non-orthogonal multiple access in 5G systems: Tractability and computation, IEEE Trans. Wireless Commun. 15 (12) (2016) 8580–8594.
[15] S. Lv, W. Zhuang, X. Wang, C. Liu, X. Hu, Y. Sun, X. Zhou, A performance study of CSMA in wireless networks with successive interference cancellation, in: 2012 IEEE International Conference on Communications, ICC, IEEE, 2012, pp. 466–471.
[16] M. Zou, S. Chan, H.L. Vu, L. Ping, Throughput improvement of 802.11 networks via randomization of transmission power levels, IEEE Trans. Veh. Technol. 65 (4) (2015) 2703–2714.
[17] A. Sankararaman, F. Baccelli, CSMA k-SIC—A class of distributed MAC protocols and their performance Evaluation, in: 2015 IEEE Conference on Computer Communications, INFOCOM, IEEE, 2015, pp. 2002–2010.
[18] M. Alvandi, M. Mehmet-Ali, J.F. Hayes, Delay optimization and cross-layer design in multihop wireless networks with network coding and successive interference cancelation, IEEE J. Sel. Areas Commun. 33 (2) (2014) 295–308.
[19] C. Jiang, Y. Shi, X. Qin, X. Yuan, Y.T. Hou, W. Lou, S. Kompella, S.F. Midkiff, Cross-layer optimization for multi-hop wireless networks with successive interference cancellation, IEEE Trans. Wireless Commun. 15 (8) (2016) 5819–5831.
[20] D. Yuan, V. Angelakis, L. Chen, E. Karipidis, E.G. Larsson, On optimal link activation with interference cancelation in wireless networking, IEEE Trans. Veh. Technol. 62 (2) (2012) 939–945.
[21] E. Karipidis, D. Yuan, Q. He, E.G. Larsson, Max–min power control in wireless networks with successive interference cancelation, IEEE Trans. Wireless Commun. 14 (11) (2015) 6269–6282.
[22] L. Qu, J. He, C. Assi, Understanding the benefits of successive interference cancellation in multi-rate multi-hop wireless networks, IEEE Trans. Commun. 62 (7) (2014) 2465–2477.
[23] C. Xu, K. Ma, Y. Xu, Y. Xu, Y. Fang, Optimal power scheduling for uplink transmissions in SIC-based industrial wireless networks with guaranteed real-time performance, IEEE Trans. Veh. Technol. 67 (4) (2017) 3216–3228.
[24] C. Xu, M. Wu, Y. Xu, Y. Fang, Uplink low-power scheduling for delay-bounded industrial wireless networks based on imperfect power-domain NOMA, IEEE Syst. J. (2019) 1–12.
[25] Y. Xu, Q. Wang, J. Liu, C. He, B. Diao, Y. Xu, PDMAC-SIC: Priority-based distributed low delay MAC with successive interference cancellation for industrial wireless networks, in: 2019 20th Asia-Pacific Network Operations and Management Symposium, APNOMS, IEEE, 2019, pp. 1–6.

[26] INRIA, WSNET, http://wsnet.gforge.inria.fr.
[27] T. Zheng, M. Gidlund, J. Åkerberg, WirArb: A new mac protocol for time critical industrial wireless sensor network applications, IEEE Sens. J. 16 (7) (2015) 2127–2139.
[28] V. Shnayder, M. Hempstead, B.-r. Chen, G.W. Allen, M. Welsh, Simulating the power consumption of large-scale sensor network applications, in: Proceedings of the 2nd International Conference on Embedded Networked Sensor Systems, 2004, pp. 188–200.
[29] D. Yang, Y. Xu, H. Wang, T. Zheng, H. Zhang, H. Zhang, M. Gidlund, Assignment of segmented slots enabling reliable real-time transmission in industrial wireless sensor networks, IEEE Trans. Ind. Electron. 62 (6) (2015) 3966–3977.



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